

TRIAPERFORMANCE GUIDE

How to fuel your long sessions

The pre-session formula, how many grams per hour, and how to train your gut to tolerate them

Four sections from the Triaperformance Fuel Kit: what to eat before, how much to take per hour by duration and sport, how to raise that number without wrecking your stomach, and what to do the day it shuts down anyway.

What this guide covers

This is training material. It is not medical advice, and it is not individual nutrition advice.

It covers one thing, and it covers it completely: **how to fuel your sessions and your races.**

It does not cover diets, weight loss, body composition or supplementation. Not because those don't matter, but because they aren't my job. I'm your coach, not your dietitian, and that distinction isn't a technicality — it's the line between what I can defend from experience and what requires a licence.

Talk to your doctor or a registered dietitian before applying any of this if you:

- Have diabetes of any type
- Have coeliac disease, irritable bowel syndrome, or another digestive condition
- Have high blood pressure or kidney disease
- Are pregnant or trying to become pregnant
- Take regular medication
- Have a history of disordered eating
- Are under 18

None of what follows is written for those situations without professional supervision.

From the coach

You train seriously. You're up before the sun, you finish the intervals even when they burn, you put in the hours on the bike. You're building an engine.

The problem is that most athletes I coach arrive with the engine well built and the fuel improvised. They rehearse pace. They rehearse cadence. They rehearse transitions. Nutrition gets decided the night before.

This guide exists to close that gap. It won't tell you how to eat in daily life. It will tell you **how much, when, and in what form to put fuel into and around your sessions**, with numbers you can use tomorrow.

One rule organises everything else: **the gut is trainable, and nothing is new on race day.**

The athlete who rehearses fuelling with the same discipline as intervals stops surviving races and starts racing them.

Let's get to work.

Iván

Founder & Head Coach, Triaperformance

1. Before the session

The rule: one gram per kilo, per hour remaining

It's the only formula worth memorising.

Carbohydrate (g) = your body mass (kg) × hours until the start

A 70 kg athlete:

Time before the start	Carbohydrate	Example
3–4 hours	210–280 g	White rice with chicken and a piece of fruit; a large bowl of oats with banana and honey
2 hours	140 g	Turkey sandwich on white bread + banana; bowl of cereal with low-fat milk
1 hour	70 g	Two slices of white bread with jam; a sports drink
30 minutes	35 g	A gel with water; half a ripe banana; a fruit pouch

The closer to the start, the less fibre and the less fat. Both slow gastric emptying, and whatever is still sitting in your stomach when the effort begins is what ruins the session.

The window worth avoiding

If you're taking carbohydrate before a hard effort, take it **more than 60 minutes out or in the final 10–15 minutes**. The middle ground — 30 to 45 minutes before — is where some athletes feel an energy dip in the first few kilometres.

It doesn't happen to everyone, and a proper warm-up blunts it. But if you've ever started a race feeling empty despite a good breakfast, that's the most likely reason, and the fix is free: move the gel.

Training fasted

It has a use, and it's a narrow one: **easy, short aerobic work**. There it's a legitimate tool.

For anything with intensity, training unfuelled will cap your pace and won't give you an adaptation you couldn't get while fed. This isn't a closed question in the literature — there are serious "train-low" protocols in use at the elite level — but for an athlete with a job, a family and 8 to 12 hours a week, the cost outweighs the return.

Simple rule: fasted on easy runs only. Everything else gets fuelled.

2. During the session

This is where the long session is won or lost. And it's where most athletes go short.

The ladder: how much per hour

Duration	Target per hour	Note
Under 45 min	Nothing	Water if it's hot.
45–75 min at high intensity	Mouth rinse, or 15–30 g	The benefit is central, not muscular. Swilling a sports drink for 5–10 seconds and spitting it out works.
1–2 hours	30–60 g	
2–3 hours	60–90 g	You'll need mixed sources. See below.
Over 3 hours	90 g, up to 120 g on the bike	Only with a trained gut.

The ceiling changes by discipline

Almost no guide separates this, and it's the difference between a plan that works and one that leaves you throwing up at 30 km.

Discipline	Practical ceiling	Why
Bike	100–120 g/h	No impact, you're seated, the stomach tolerates more.
Run	60 g/h baseline, up to 90 g/h with a trained gut	Impact and posture cut tolerance.
Swim	Not applicable	You can't eat while swimming. It gets solved before you get in.

Start early. At 20–30 minutes, not when you feel hungry. By the time you're hungry in a long session you're already behind, and clawing it back on a compromised stomach rarely works.

Above 60 g/h you need two sugars

Your gut absorbs glucose through one transporter and fructose through another. The glucose route saturates near 60 g/h. Push past that with a single sugar and the excess sits in your gut and causes trouble.

Look for products combining glucose (or maltodextrin) and fructose at a ratio near 1:0.8.

It's on the label of any modern product. With a single sugar, your real ceiling is 60 g/h no matter how much you train.

What to use

Context	Options
Running, or high intensity	Gels, chews, sports drink. Immediate digestion.
Bike, steady intensity	All of the above, plus bars, banana, boiled potato with salt, pressed rice cakes.
Last 20 min before running	Liquid only. Empty the stomach before you get off the bike.

Count in grams, not in units. A gel isn't a measurement: they range from 20 g to 40 g depending on brand. Read the label once, write the number down, and build your plan in grams. "Three gels an hour" means nothing; "75 grams an hour" means something.

3. Train your gut

This is the most important section in the guide.

You can know exactly how many grams you need and still be unable to absorb them, because your gut's capacity to process carbohydrate during exercise **is trainable, just like the engine**. An athlete who has never exceeded 40 g/h in training won't tolerate 90 g/h on race day. They'll tolerate nausea.

One honest caveat: the evidence on the exact mechanism is thinner than it's usually presented, and the studies aren't entirely consistent. What is consistent — and anyone who coaches long-distance athletes will confirm it — is that **repeated practice reduces symptoms and raises the amount you tolerate**. That's enough to build a protocol on.

The 6-week block

Run it **only on the week's long or race-specific session**. Never on a recovery run: there it does nothing useful and leaves you heavy for no return.

Weeks	Target per hour	How
1–2	30–40 g/h	One product only. The one you'll race with.
3–4	60 g/h	Add the second source (glucose + fructose).
5–6	80–90 g/h running · 100–120 g/h on the bike	At race intensity, with race products.

The four rules

1. **One variable at a time.** If you raise the amount, don't change the brand that week.
2. **Score symptoms 1 to 10** after every long session. Bloating, nausea, cramping, urgency.
3. **Above 5, you hold that dose another week.** You don't advance. The block stretches; it doesn't get forced.
4. **At race intensity, not easy pace.** Tolerating 90 g/h on a jog predicts nothing about tolerating it at marathon pace. The blood doing the digesting is the blood moving your legs.

When to back off

Vomiting, cramping that forces you to stop, or diarrhoea mid-session: **drop one step and stay there two weeks**. That isn't the plan failing, it's information. Today's ceiling is one rung lower than the one you tested.

And the rule that contains all of it

Nothing new on race day. Not a brand, not a flavour, not a concentration, not a breakfast time. If you're racing an Ironman on one brand of gel, you train on it for months first.

4. When your stomach shuts down

It will happen. The question isn't whether, it's what you do.

Diagnosis, after the session

If you finish with diarrhoea or gut discomfort, there are two likely causes and they're easy to tell apart:

1. **Product tolerance.** You usually already know: one brand or flavour always disagrees with you. Swap it and move on.
2. **Concentration versus intake rate.** The common one, and the one nobody calculates. **Recompute the grams against the minutes you actually consumed them in**, not against the full hour. If you took 90 g in 20 minutes, your gut didn't receive 90 g/h — it received the equivalent of 270 g/h during that window. The fix is to dilute more, or to spread it better.

Mid-race

When the stomach shuts down, in this order:

1. **Slow down.** It's the only thing that returns blood flow to the gut. Non-negotiable, and first.
2. **Drop the solids.** Move to liquid.
3. **Dilute.** More water for the same carbohydrate.
4. **Cut the caffeine** for the rest of the race.
5. **Walk the aid stations.** Swallowing while running makes everything worse.
6. **Lower your hourly target** and accept less. Finishing on 40 g/h beats abandoning on 90.

Once it settles, build back gradually. Don't try to recover what you missed: it doesn't come back, and the attempt is what causes the second episode.

And what I tell my athletes: will it happen in the race? Yes. That's why we train it.

This is a quarter of the full guide

The complete Fuel Kit adds hydration and sodium, caffeine, carbohydrate loading, heat and altitude, and a race template for nine distances. It is inside Triaperformance All-Access, along with the rest of the library.

See All-Access →

Would you rather we fitted it to your actual race? Write to me. coach@triaperformance.com · [WhatsApp](#)